

GAME DESIGN DOCUMENT (GDD)

Game Name: Pebble Promise

Genre: Adventure, Narrative

Game Elements: Players can talk with certain villagers, interact with objects, and explore the village.

Player: Single-player only

Technical Specs

Technical Form: 3D graphics

View: Third-person

Platform: PC (Windows)

Language: Blueprints, developed in Unreal Engine 5.7.1

Device: PC

Gameplay

The game is a short narrative driven adventure game focused on exploration, dialogue and simple quest mechanics. The player is a penguin living in a small village. He needs to collect perfect pebbles, to get closer to his final goal - his loved one Marika. The game ends with gifting a bracelet to her.

Game Play Outline

Opening the game application: Launch screen, then the main menu is opened

Game options: New Game, Quit

Modes: Story mode only

Game elements: dialogues with NPCs, collectible stones, quests, interactable doors

Game levels: Village (main one), and interior locations like Inn, Blacksmith, Player House (only at the beginning of the game)

Player's controls: WASD to move, mouse to look, space to jump, E to interact

Winning: Player collects all needed stones, completes the blacksmith quest, receives a bracelet and gives it to NPC Marika.

Losing: No traditional losing condition, but the player may get stuck if he can't find required stones

End: After giving a bracelet to Marika, the game displays an ending screen and returns to the main menu.

Why is all this fun?

The game offers a calm and cozy atmosphere, simple but emotional storytelling and a sense of purpose and progression. Finding stones is relaxing and rewarding. Players can enjoy a simple, short story instead of trying to figure out simple mechanics for 3 hours.

Key Features

- story driven gameplay
- dialogue systems
- quests
- collectible items

GameObject behaviours

- NPCs: dialogue trees
- Stones: collectible items, tracked by UI
- Doors: trigger level change
- bracelet: quest reward

Design Guidelines

All stones must be reachable. The game avoids stress-based mechanics. The environment is designed to enhance the experience while remaining visually neutral, ensuring the player's attention to the story.

Game Design Definitions

- **Win:** Completing the story by giving the bracelet to Marika.
- **Loose:** Player gets stuck, can't find stones.
- **Level transition:** When the player enters houses, interacts with doors.
- **Main gameplay focus:** Exploration, dialogue, simple quests.
- **Saving:** -

Player Definition

Player controls a penguin character exploring a small village. The Character has no name and acts as a vessel for the player to experience the story.

Player Properties

- collected stones count
- quests progress statuses
- bracelet possession status

Player Rewards

- stones, used for crafting a bracelet
- bracelet, a quest reward
- narrative progression

Player Definitions

- **Move:** navigate 3D environment using WASD
- **Look around:** mouse to look around
- **Interact:** use E to interact, collect stones, enter houses, talk to NPCs

User Interface (UI)

The UI is minimalistic and clean. Designed to keep the players focused on the game itself. Key elements are positioned at the edge of the screen to minimize distraction.

Main UI elements:

- collected stones counter
- Interaction prompts
- dialogue windows
- Main Menu and END screen

Assets

<https://sketchfab.com/3d-models/penguin-2c079bc491fb4bb4942c0f87927f8d87>

<https://www.fab.com/listings/52529a12-e88e-41a0-8834-b87306f20c24>

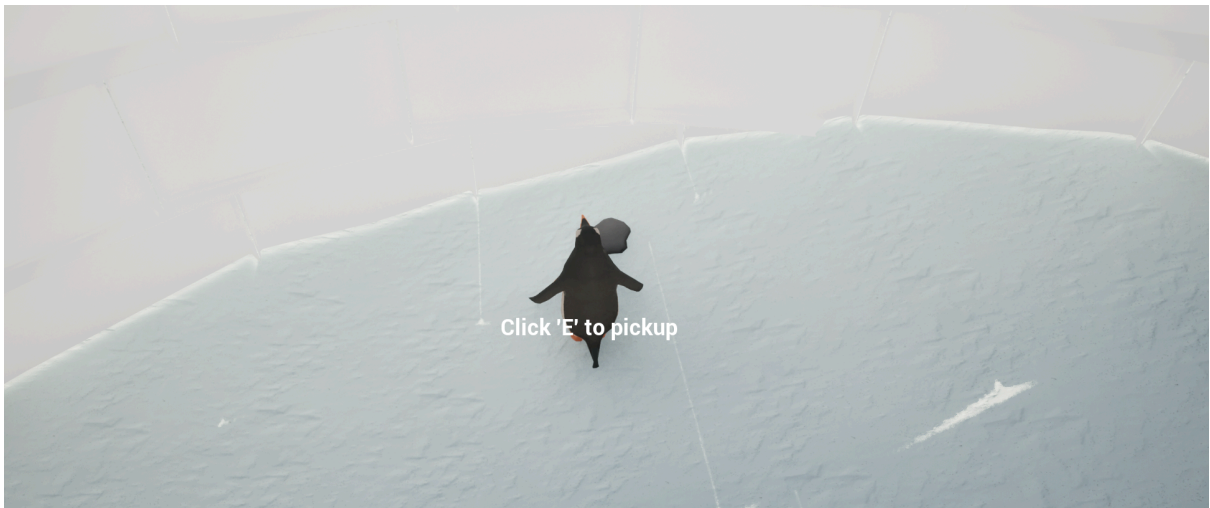
<https://www.fab.com/listings/7d647b9c-850f-4453-9d58-abdbdb3d8675>

<https://www.fab.com/listings/766c8bc1-7922-42cd-b483-e266c2d753a6>

<https://www.fab.com/pl/listings/6ac5c478-8f2c-47cc-88e1-ac36aac010c8>

Game Screenshots





Repository

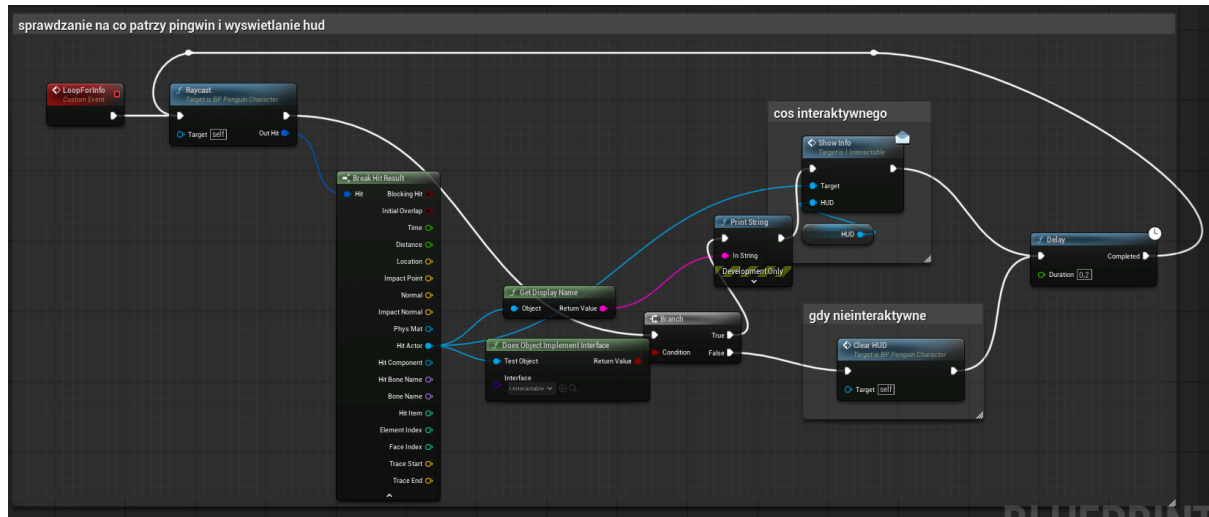
<https://github.com/Lesiaaa/PebblePromise>

Used tools and websites

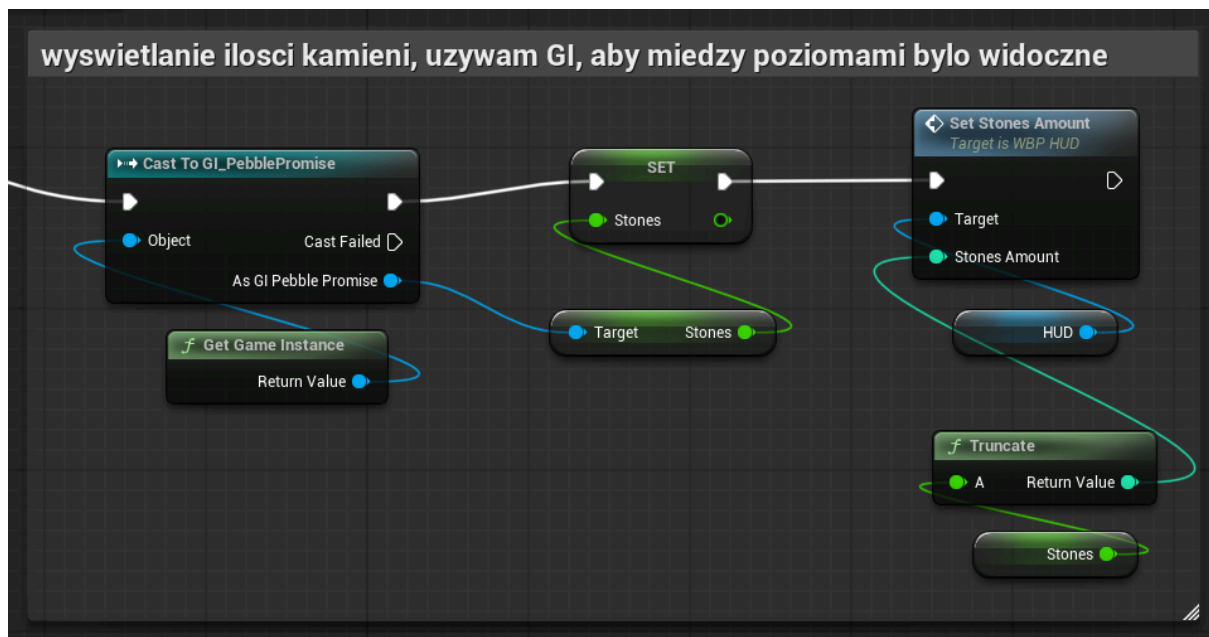
- <https://dev.epicgames.com/documentation/en-us/unreal-engine/unreal-engine-5-7-documentation>
- <https://www.youtube.com/watch?v=xH2Wlr2RNhw&t=4s>
- <https://www.youtube.com/watch?v=ujFVHCbnc0Q&t=225s>
- <https://www.youtube.com/watch?v=LF8yKckkero>
- <https://dev.epicgames.com/documentation/en-us/unreal-engine/saving-and-loading-your-game-in-unreal-engine>
- <https://www.youtube.com/@UnrealEngine/courses>
- chat gpt, for helping with learning Unreal Engine, refining code and tricky bugs :))

Code important parts explained

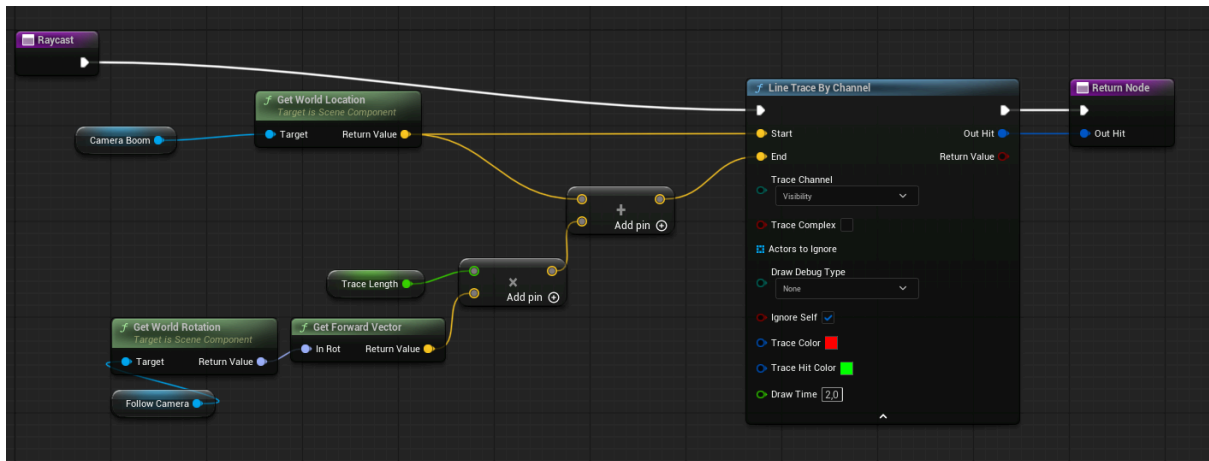
Checking what the player character is looking at using raycast. If raycast hits an object and if the object is interactable, an interaction prompt is displayed on the HUD.



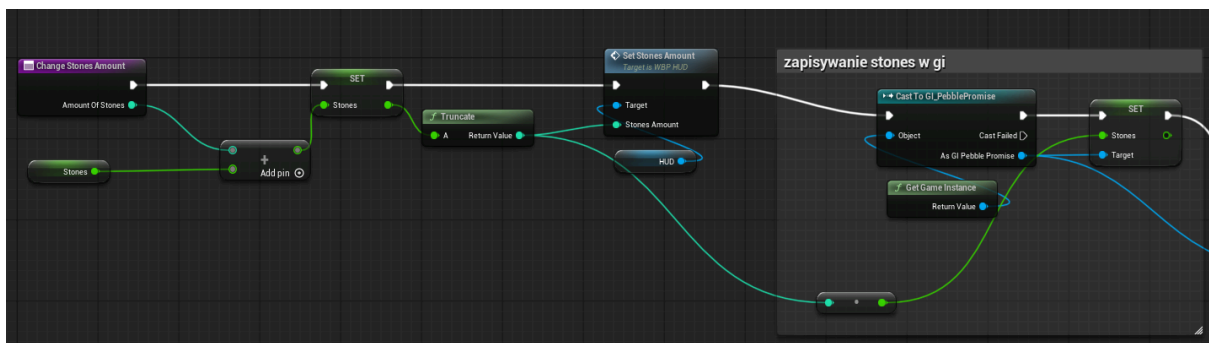
Checking the current state of the stones. We use GI_PebblePromise file to store the stones amount.



The Raycast function starts at the camera's world location and extends forward in the direction the camera is facing, based on trace length. If the raycast hits an object on the visibility channel, information about hit is returned.



Changing the current stones amount. The new value is calculated and stored and then updated to HUD.



Ideas for future development

- additional levels, island
- travel system between different places where penguins live via wooden signs
- mini-games/levels, ex. jumping on ice floes to reach the other side, avoiding polar bears
- shop, where player can buy for stones different items to help him in different levels/mini games
- more dialogues with villagers (NPCs)

Inspirations and Ideas

- Emotional storytelling inspired by games such as 'Ori and the Will of the Whisps'
- Short, relaxing games like Journey